

INTERNSHIP ASSIGNMENT

TASK 3 - CAUSALITY ASSESSMENT

ADR Reporting and Documentation

Student Name	Md Sadumar
Roll No.	B.Pharm Internship – Task 3
Course / Program	B.Pharm
College / University	Jagannath University, Jaipur
Hospital / Organization	Pharmacovigilance Department
Internship Duration	Pharmacy Internship Training
Submission Date	15 September 2026

Assignment ID: TASK-3 | Status: COMPLETED

1. Introduction

Causality assessment is the systematic evaluation of the relationship between a medicinal product and a suspected adverse drug reaction (ADR). It examines how strongly available clinical evidence supports an association between the medicine and the event. It is an important part of pharmacovigilance because it improves the quality and interpretation of individual case safety reports.

The assessment considers timing, dechallenge, rechallenge when it occurs, alternative causes, known pharmacology, dose-response and objective clinical evidence.

2. Objectives

- Understand the concept and purpose of causality assessment.
- Evaluate the relationship between a suspected medicine and an adverse reaction.
- Apply standard causality assessment methods to case studies.
- Classify reactions as Certain, Probable/Likely, Possible and other applicable categories.
- Understand the importance of complete and accurate ADR documentation.

3. Basic Principles of Causality Assessment

- Temporal relationship – the event occurs after exposure within a biologically plausible period.
- Dechallenge – the reaction improves after dose reduction or discontinuation.
- Rechallenge – the reaction recurs after re-exposure; it can be strong evidence but should not be performed solely for testing.
- Alternative causes – other medicines, diseases, infections, foods or exposures may explain the event.
- Known association – the reaction is consistent with established clinical or pharmacological knowledge.
- Objective evidence – laboratory, imaging or examination findings support the reaction.
- Dose-response – changes in dose or exposure correspond with changes in the event.

4. Standard Causality Assessment Methods

The WHO-UMC causality system and Naranjo algorithm are commonly taught approaches. The selected method should follow the institutional or pharmacovigilance procedure.

Method	Main features	Output
WHO-UMC	Clinical judgment using timing, alternative causes, dechallenge/rechallenge and known drug-event relationship.	Certain; Probable/Likely; Possible; Unlikely; Conditional/Unclassified; Unassessable/Unclassifiable.
Naranjo	Structured questions with scores for timing, dechallenge, rechallenge, alternatives and objective evidence.	Definite; Probable; Possible; Doubtful.

5. WHO-UMC Causality Categories

Category	Interpretation
Certain	Plausible time relationship; no convincing alternative cause; clinically reasonable withdrawal response and, where appropriate, definitive re-exposure evidence.
Probable / Likely	Reasonable time relationship; alternative causes unlikely; reasonable response after withdrawal; rechallenge not required.
Possible	Reasonable time relationship, but disease or other medicines could also explain the event.
Unlikely	Timing or other evidence makes a drug relationship improbable.

Category	Interpretation
Conditional / Unclassified	More information is needed or awaited.
Unassessable / Unclassifiable	Information is insufficient or contradictory.

6. Naranjo Algorithm - Completed Sample

For the illustrative antibiotic-rash case, the following completed assessment gives a sample score of 6, interpreted as Probable. This is an academic case for internship documentation.

Question	Completed answer	Score
Event after suspected drug?	Yes – rash began 2 days after antibiotic.	+2
Improved after discontinuation?	Yes – improved after stopping drug.	+1
Reappeared on readministration?	No rechallenge performed.	0
Alternative causes?	No convincing alternative cause identified.	+2
Reaction with placebo?	Not observed / not applicable.	0
Toxic concentration?	Not documented.	0
Dose-response?	Not demonstrated.	0
Previous similar reaction?	Not documented.	0
Objective evidence?	Clinical examination documented rash.	+1
TOTAL	Naranjo score = 6	PROBABLE

7. Case Study 1 - Completed

Patient: 45-year-old adult. Drug: Amoxicillin 500 mg orally three times daily. Indication: Bacterial infection. Reaction: Generalized itchy maculopapular rash on Day 3. Dechallenge: Positive; drug stopped and rash improved over 4 days. Rechallenge: Not performed. Alternative causes: None convincing. WHO-UMC: Probable/Likely. Naranjo: 6 – Probable.

8. Case Study 2 - Completed

Patient: 30-year-old adult. Drug: New antihypertensive. Reaction: Headache beginning 3 days after treatment. Other factors: Poor sleep and dehydration. Dechallenge: Not available because treatment continued. Outcome: Symptoms fluctuated. Assessment: Possible, because alternative causes are plausible.

9. Case Study 3 - Completed

Patient: Adult with no previous history of the reaction. Drug: Medicine previously associated with the same reaction. Event: Characteristic reaction soon after initiation. Dechallenge: Positive; reaction resolved. Rechallenge: Positive during later clinically necessary re-exposure. Alternative causes: None identified. Assessment: Certain, provided the complete record supports it.

10. Documentation Required for Causality Assessment

- Patient age/sex and relevant medical history.
- Medicine name, dose, route, frequency, indication and dates.
- Reaction description and onset date/time.
- Concomitant medicines and other relevant exposures.
- Laboratory, imaging and examination findings.
- Action taken, dechallenge and outcome.
- Rechallenge information if it occurred naturally.
- Alternative causes considered.
- Reporter and follow-up details.

11. Practical ADR Causality Assessment - Filled Form

Field	Completed Entry
Patient / Case ID	ADR-CASE-2026-003
Student / Reporter	Md Sadumar
Course	B.Pharm
College / University	Jagannath University, Jaipur
Patient	45-year-old adult
Suspected medicine	Amoxicillin 500 mg tablet, oral, three times daily
Indication	Bacterial infection
Adverse reaction	Generalized itchy maculopapular rash
Time to onset	Approximately 2 days after starting therapy
Dechallenge	Yes – medicine discontinued; rash improved over 4 days
Rechallenge	No – not performed for safety reasons
Alternative causes	No convincing alternative cause identified
Investigations	Clinical examination documented generalized rash
Severity	Moderate
Seriousness	Non-serious; no hospitalization or life-threatening outcome
Management	Medicine discontinued; symptomatic treatment provided
Outcome	Recovered / clinically improved
Causality method	WHO-UMC + Naranjo algorithm
WHO-UMC category	Probable / Likely
Naranjo score	6 – Probable
Assessor	Md Sadumar – B.Pharm Intern
Assessment date	15 September 2026
Follow-up status	Completed – academic case documentation

12. Conclusion

Causality assessment is a structured and clinically informed process for evaluating the relationship between a medicine and an adverse reaction. Accurate chronology, clinical information, alternative causes, dechallenge and rechallenge evidence are essential. WHO-UMC and Naranjo methods support assessment but do not replace professional clinical judgment. Complete documentation and follow-up strengthen pharmacovigilance and patient safety.

13. References

1. World Health Organization (WHO). Pharmacovigilance and adverse drug reaction monitoring resources.
2. Uppsala Monitoring Centre (UMC). WHO-UMC system for standardised case causality assessment.
3. Naranjo CA, Busto U, Sellers EM, et al. A method for estimating the probability of adverse drug reactions. Clinical Pharmacology & Therapeutics. 1981.
4. Indian Pharmacopoeia Commission (IPC). Pharmacovigilance Programme of India (PvPI) resources and ADR reporting guidance.
5. Relevant institutional SOPs and current official pharmacovigilance reporting procedures.

Submission Status: COMPLETE – all assignment fields and assessment columns are filled.

Student	Supervisor
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Md Sadumar	Pharmacovigilance Supervisor
15 September 2026	15 September 2026